

- 2 (a) *either* the half-life of the source is very long
or decay constant is very small
or half-life $\gg 40$ days
or decay constant $\ll 0.02 \text{ day}^{-1}$ B1 [1]

(b) number of helium atoms = $3.5 \times 10^6 \times 40 \times 24 \times 3600$ C1
 $= 1.21 \times 10^{13}$

either $pV = NkT$ *or* $pV = nRT$ and $n = N / N_A$ C1

$1.5 \times 10^5 \times V = 1.21 \times 10^{13} \times 1.38 \times 10^{-23} \times 290$

$V = 3.2 \times 10^{-13} \text{ m}^3$ A1 [3]

(if uses $T/^{\circ}\text{C}$ or $n = 1$ or $n = 4$, then 1 mark max for calculation of number of atoms)

- 3 (a) increasing separation of molecules / breaking bonds between molecules B1